Writing the methods section

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Write methods that expose the decisions affecting the evidence

Specify exact inputs, boundaries, data roles and versions while keeping unassessed design details explicit.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Write methods that expose the decisions affecting the evidence

> Specify exact inputs, boundaries, data roles and versions while keeping unassessed design details explicit.

Learning goals

I can explain the **mechanism** and terminology of writing the **methods section**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Figures that preserve **evidence**. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain writing the **methods section** using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of writing the **methods section**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Specify exact inputs, boundaries, data roles and versions while keeping unassessed design details explicit.

Start with the research problem

"We used AI and got good scores" leaves the reader guessing which model, inputs, split and scoring rule produced the result. A useful methods section supplies the decisions needed to reconstruct the design and computation. Detail matters when it changes the evidence: which sources cross a split, how text is preprocessed, what counts as success and how missing results are handled.

Do not invent those details when auditing a report that omits them.

Original concept explanation

A **methods section** should let a reader reconstruct the design and computation. Detail should explain choices that affect **evidence**.

Complete case inputs, rules and limits

Methods item	Complete supplied exercise contract
Main description	Used AI on data and got good scores; code and split absent
Missing main identifiers	Model/procedure version, data identity, units/splits, baseline, metric and scoring
Other required design reporting	Sampling, permissions, data roles, uncertainty assumptions, exclusions and deviations
Separate toy inputs	Exact strings oak, pine, willow
Original toy rule	Length at least four characters means long; otherwise short
Preprocessing	No whitespace trimming; supplied strings are ASCII
Changed rule	Strictly greater than four
Scope	Repeatable toy computation, not a learned model or biological classification

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Complete main-case evidence ledger

Main report omission	Why it affects assessment	What the audit should request
Model/procedure version	Which system produced the result is unknown	Exact version/configuration and code
Data identity and split	Cases, roles and source overlap unknown	Identifiers, eligibility and grouping/split rule
Baseline	Comparison target unknown	Baseline procedure and matched conditions
Metric and scoring	Good scores has no defined denominator/rubric	Explicit outcome, calculation and raw counts
Missing/exclusion handling	Reported cases may be selected	Rules, excluded/unassessed counts and deviations
Validity conclusion	Cannot assess from this description	Record missing evidence rather than fabricate a complete method

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Writing the methods section — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Methods says used AI on data and got good scores; code and split are absent.	Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.	It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Change the rule to length greater than 4.

▼ Check after predicting

Prediction: Pine becomes short; willow stays long and oak stays short.

Reason: The equality case distinguishes the two procedures.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Methods says used AI on data and got good scores; code and split are absent.

It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

Inputs: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short. No whitespace trimming occurs.

The separate toy method supplies exact strings and a rule, not a learned model. With no trimming, oak has three characters and is short; pine has four and is long; willow has six and is long under length at least four. Changing the condition to strictly greater than four changes only pine to short.

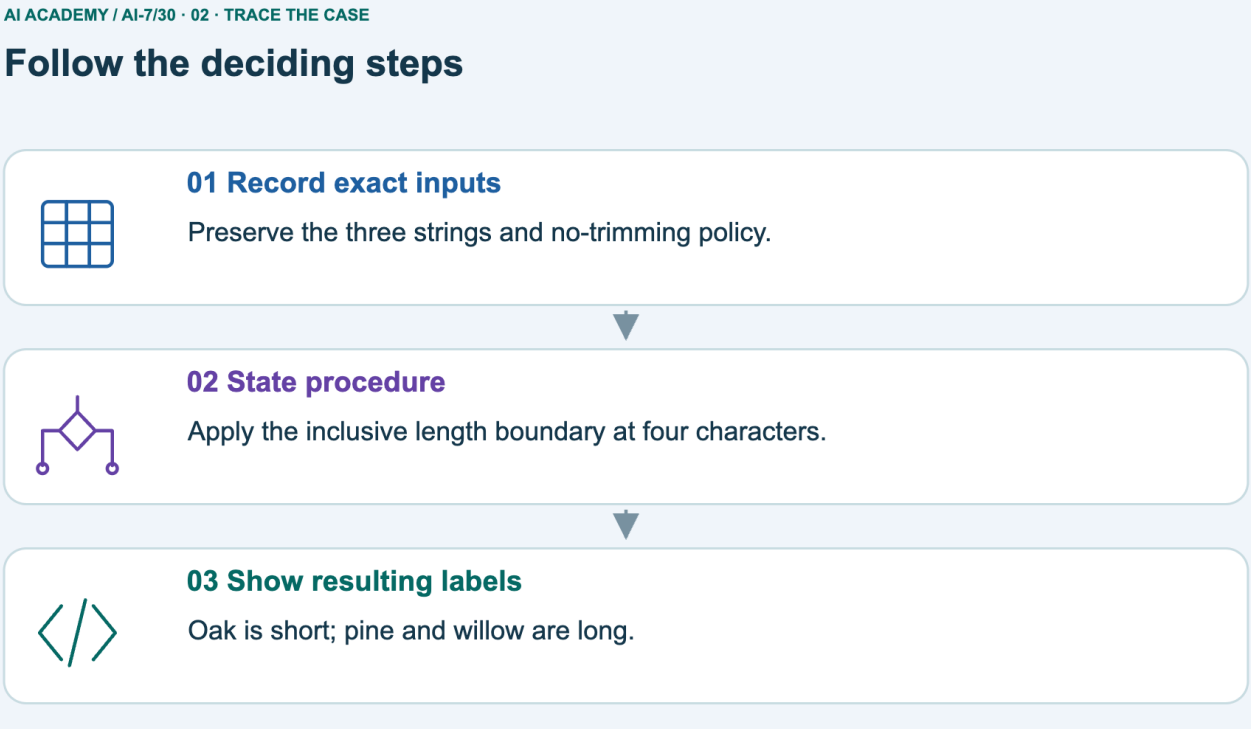
The whitespace probe oak followed by one space has four characters under this method and therefore is long; silently stripping it would change the procedure. Long and short are arbitrary string-length categories, not ecological properties. Repeating this computation does not fill in the main report’s missing AI materials.

Read the labelled representation

Method details decide the boundary case

Exact input	Characters	Length ≥ 4	Length > 4
oak	3	short	short
pine	4	long	short ← changes
willow	6	long	long

No whitespace trimming. The rule, input and inclusive/strict boundary must be recorded.
“Long” is a toy string label, not an ecological category or a learned-model result.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record exact inputs	Preserve the three strings and no-trimming policy.
2. State procedure	Apply the inclusive length boundary at four characters.
3. Show resulting labels	Oak is short; pine and willow are long.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Complete picture evidence and limits

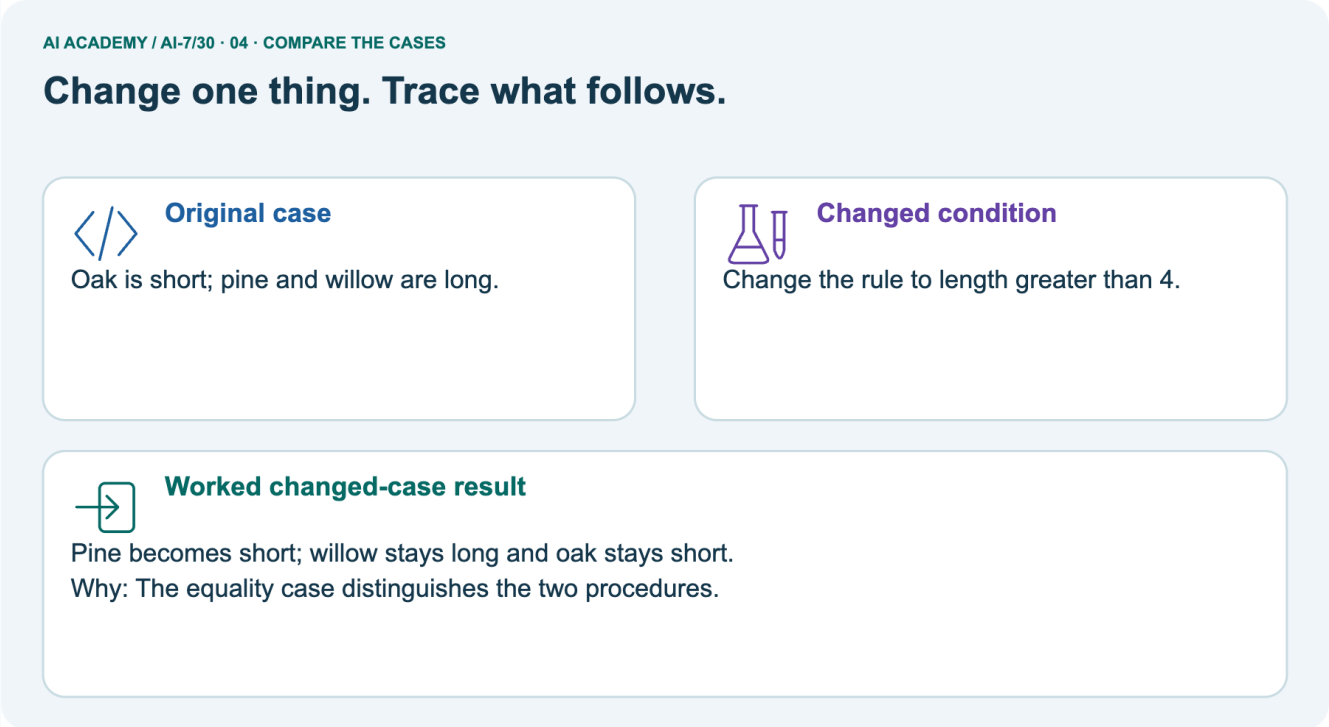
Exact toy input	Character count	Length ≥ 4	Length > 4
oak	3	short	short
pine	4	long	short
willow	6	long	long
oak followed by one space	4, because no trimming	long	short

Exact toy input	Character count	Length ≥ 4	Length > 4
Why labels differ	Boundary or preprocessing changed	Record exact method	Preserve original and revised versions

Predict the changed condition

- Change one thing: Change the rule to length greater than 4. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- Repeating this deterministic toy rule does not validate a **learned model** or establish ecological categories.

- Read the labelled picture: Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

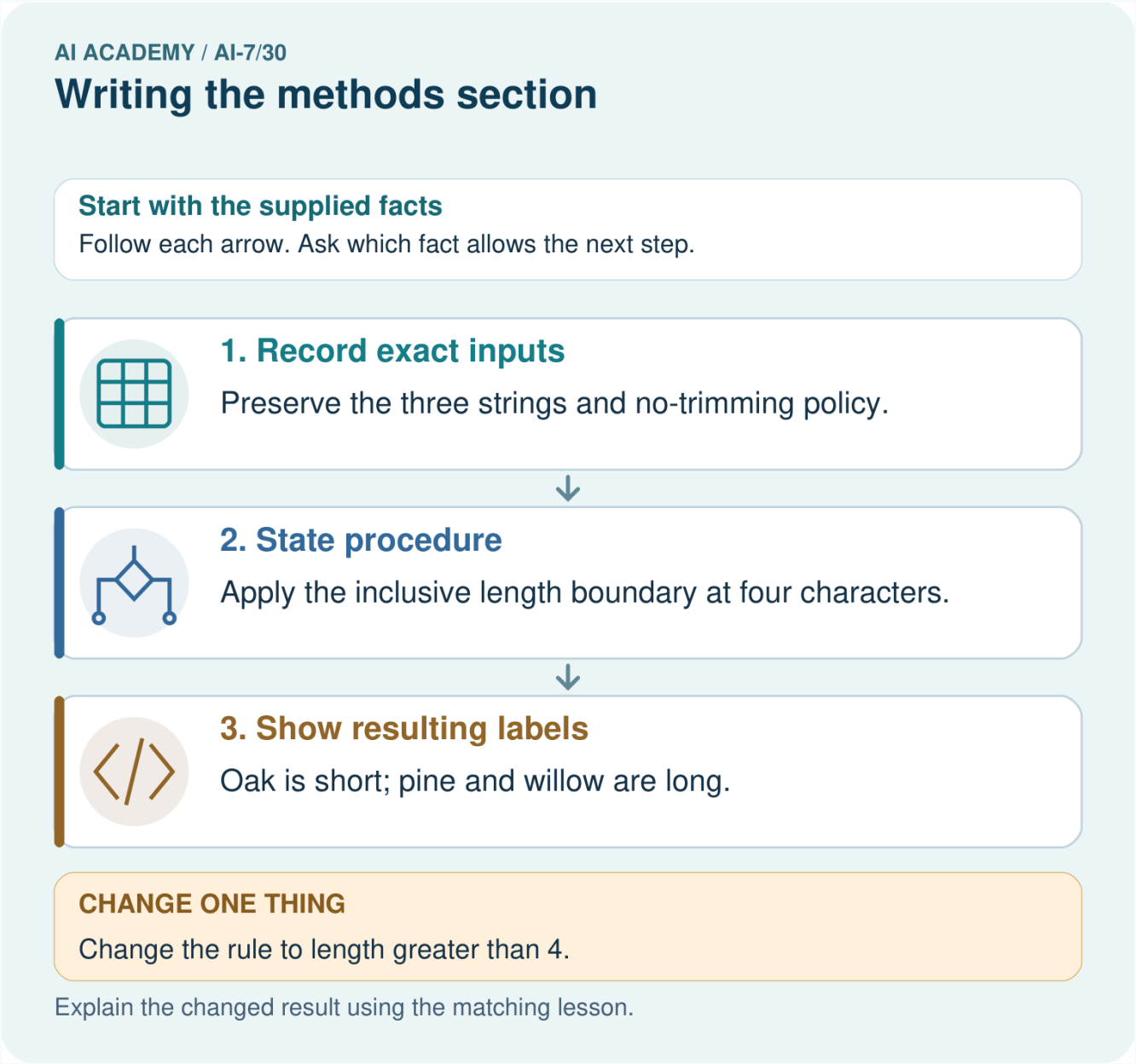
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Writing the methods section

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

A detailed **methods section** can reveal limitations; it does not guarantee a correct design.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Methods audit	Checked interpretation
Dataset named	Still need units, eligibility, roles and split/grouping rule
Metric named	Still need scoring boundary, denominator and missing handling
Near-duplicate sources across split	Can undermine evaluation independence; inspect group identities and allocation
At least four versus greater than four	Different method; pine is the boundary case
Whitespace trimming added silently	Changes the stated procedure
Code reproduces toy labels	Not evidence about a learned model
A long methods checklist is filled	Does not prove records are true or assumptions valid

Methods audit	Checked interpretation
Detail absent	Mark unknown; request evidence rather than guessing

Write and justify the repair

- A report omits this qualification: A detailed **methods section** can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example “Writing the **methods section**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Writing the **methods section**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Audit the vague AI report on paper and list the missing identifiers, data roles and scoring rules. Separately predict all three toy labels, then run or trace the full local length-rule code. Check the inclusive/strict boundary and the explicitly supplied trailing-space probe without changing the recorded input. Save original and changed outputs with their rule definitions.

   This calculation does not train a model, evaluate botanical categories or reconstruct the absent AI experiment. Keep the independent near-duplicate split scenario for a first attempt.

```
"""Repeat an explicitly specified ASCII string-length rule; no trained model."""

def labels(strings,strict=False):
 if not isinstance(strings,(list,tuple)) or any(not isinstance(s,str) for s in strings) or
type(strict) is not bool:
  raise ValueError('string sequence and Boolean boundary flag required')
 return [{'input':s,'character_count':len(s),'label':'long' if (len(s)>4 if strict else
len(s)>=4) else 'short'} for s in strings]
inputs=['oak','pine','willow'];original=labels(inputs);changed=labels(inputs,True)
assert [r['character_count'] for r in original]==[3,4,6]
assert [r['label'] for r in original]==['short','long','long']
assert [r['label'] for r in changed]==['short','short','long']
assert labels(['oak '])[0]['label']=='long' # No whitespace trimming under this method.
assert inputs==['oak','pine','willow']
print('Original explicit inclusive length rule:',original)
print('Changed strict greater-than-four rule:',changed)
print('Whitespace boundary probe:',labels(['oak ']))
print('Method: Python len on the supplied ASCII strings, no stripping or preprocessing.
Long is an arbitrary string-length label, not botanical meaning.')
print('This demonstrates repeatable computation only. It supplies no missing
```

```
code/data/split for the main AI report, no learned-model evidence and no independent study
reproduction.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Identify the grouping and split-allocation details needed to assess whether near-duplicate

sources cross evaluation roles.

Complete independent task

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled writing the **methods section evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing the **methods section**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of writing the **methods section**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Methods section	The record of study design and computation
Design choice	A decision affecting what evidence means
Analysis assumption	A condition required by a statistical procedure

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A detailed **methods section** can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Writing the methods section — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Methods says used AI on data and got good scores; code and split are absent.	Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.	It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

By the end, I can	How I will show it
Explain the mechanism and terminology of writing the methods section	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.
- Which labelled step matches this detail? Oak is short; pine and willow are long.
- 1. State procedure
- 2. Record exact **input**s
- 3. Show resulting labels
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Show resulting labels Oak is short; pine and willow are long. A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring.

Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Changed-case question and optional hint

- Changed-case practice: Change the rule to length greater than 4. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Pine becomes short; willow stays long and oak stays short.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Pine becomes short; willow stays long and oak stays short. The equality case distinguishes the two procedures.

Repair and retain the record

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

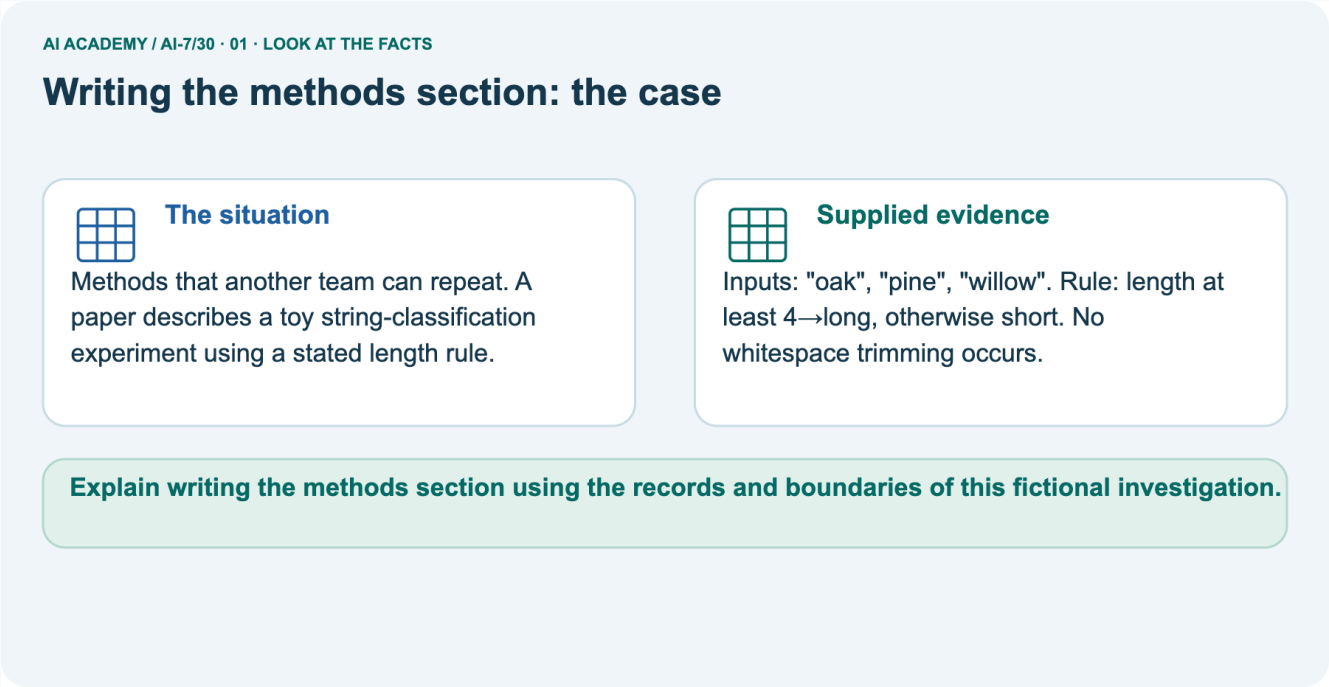
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Writing the methods section: the case



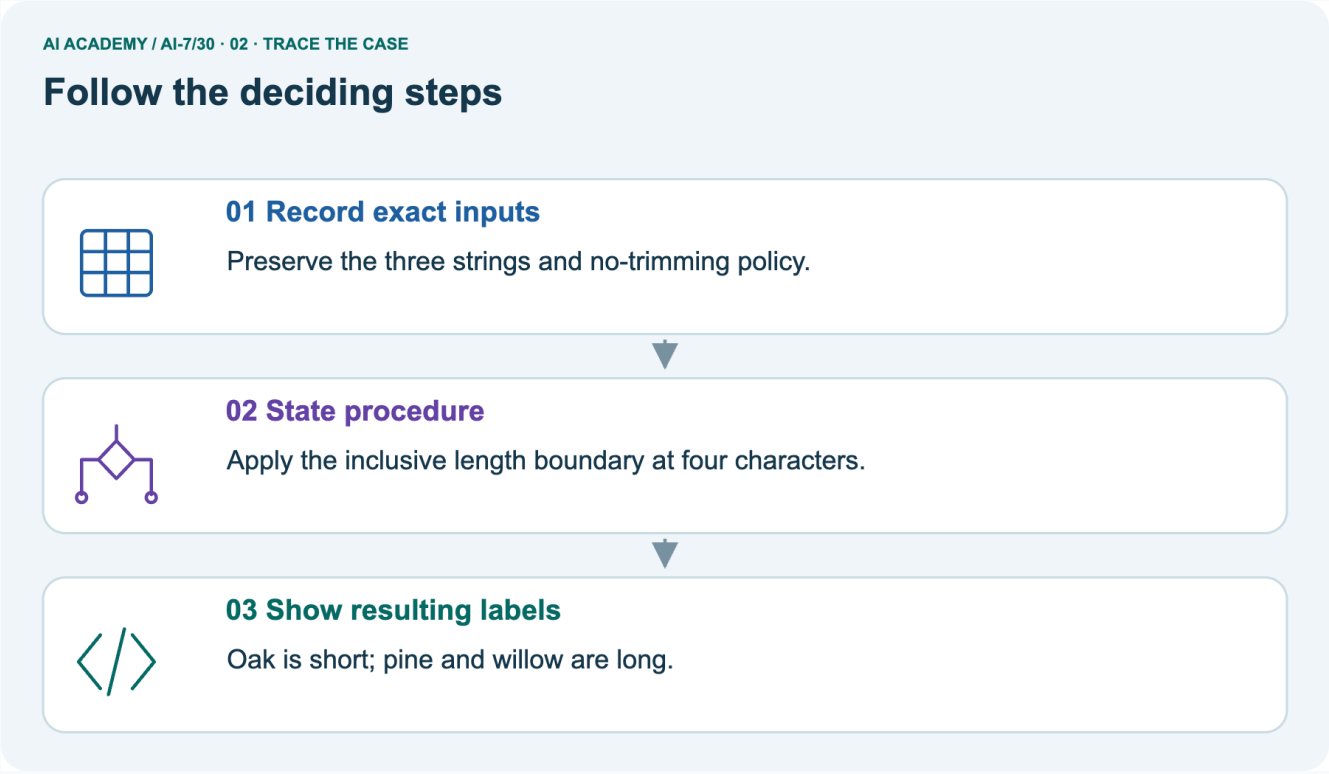
Follow the labelled picture with the matching explanation. Open the vector diagram

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule. Supplied facts: **Input**s: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short.

No whitespace trimming occurs.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

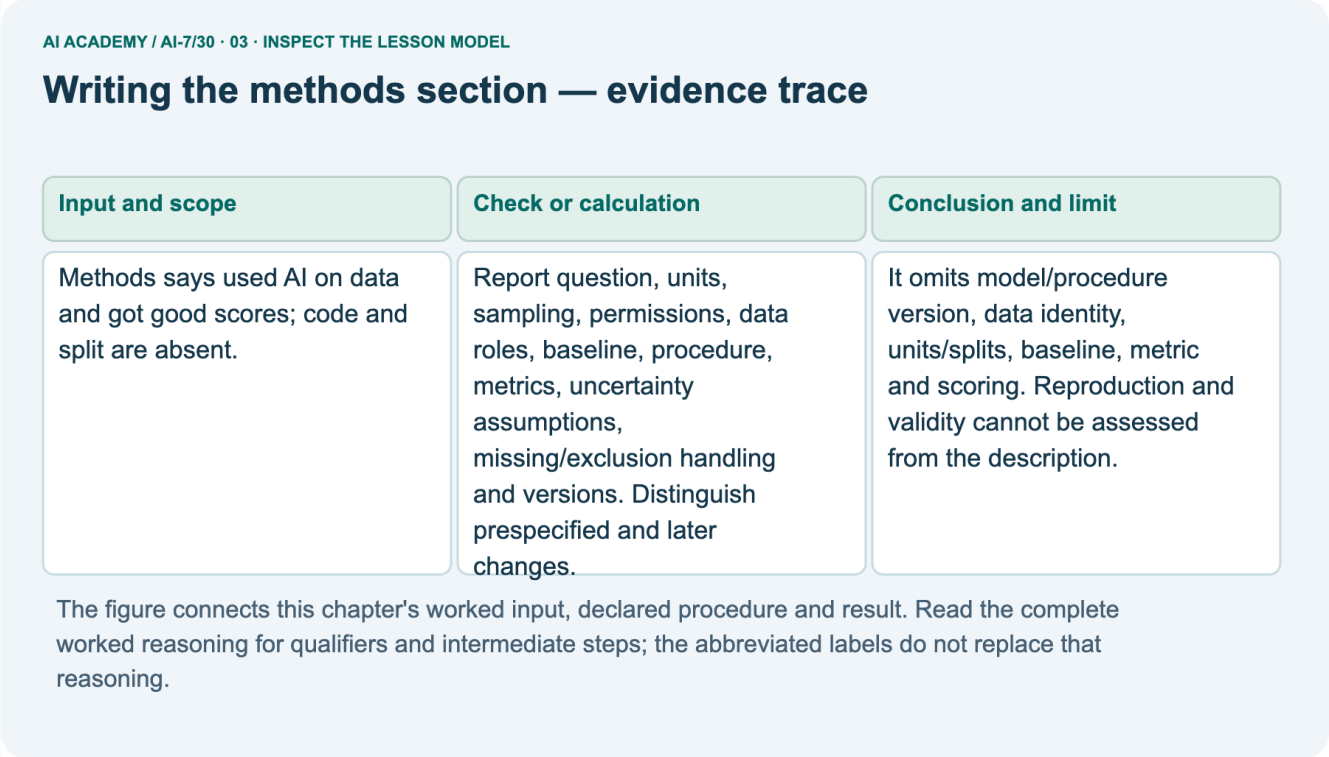


Follow the labelled picture with the matching explanation. Open the vector diagram

Record exact **input**s: Preserve the three strings and no-trimming policy. → State procedure: Apply the inclusive length boundary at four characters. → Show resulting labels: Oak is short; pine and willow are long.

Point to each numbered stage and explain which recorded **input** it uses.

03. Writing the methods section — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

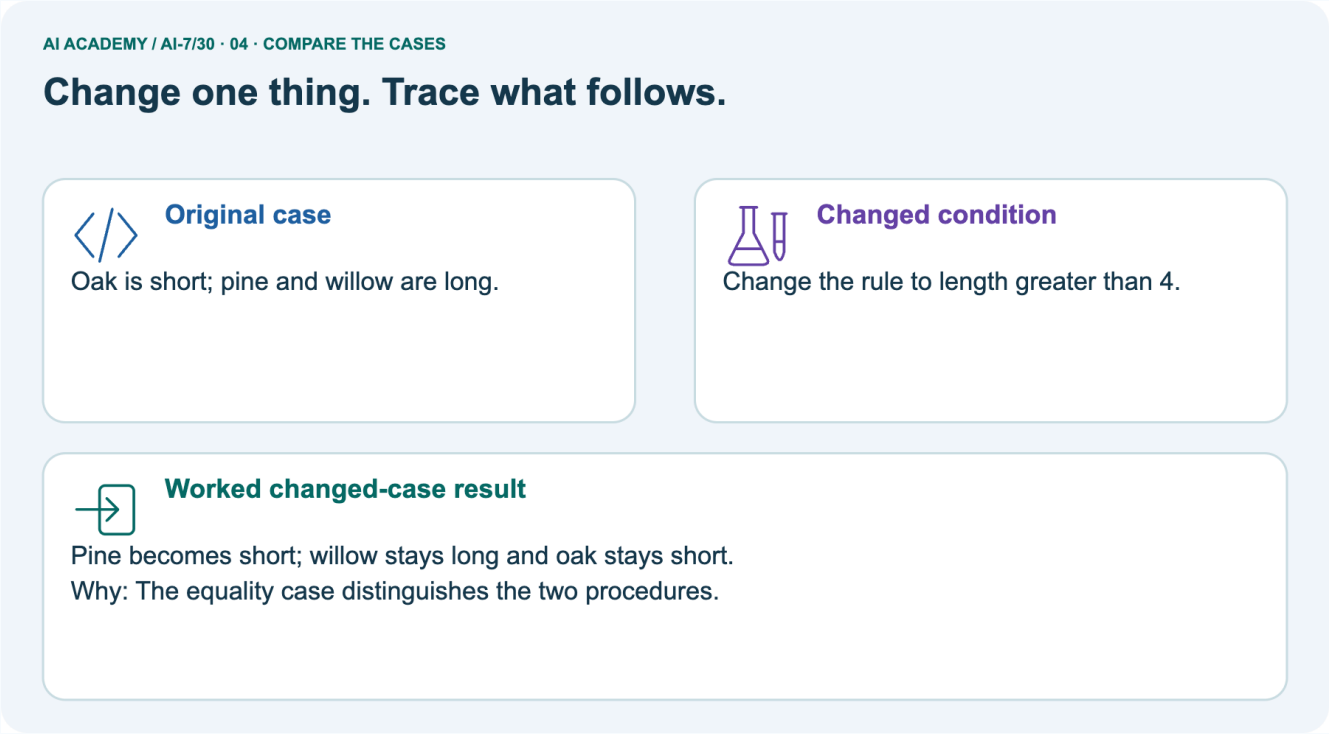
Writing the **methods section** — **evidence** trace.

Methods says used AI on data and got good scores; code and split are absent. | Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions.

Distinguish prespecified and later changes. | It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



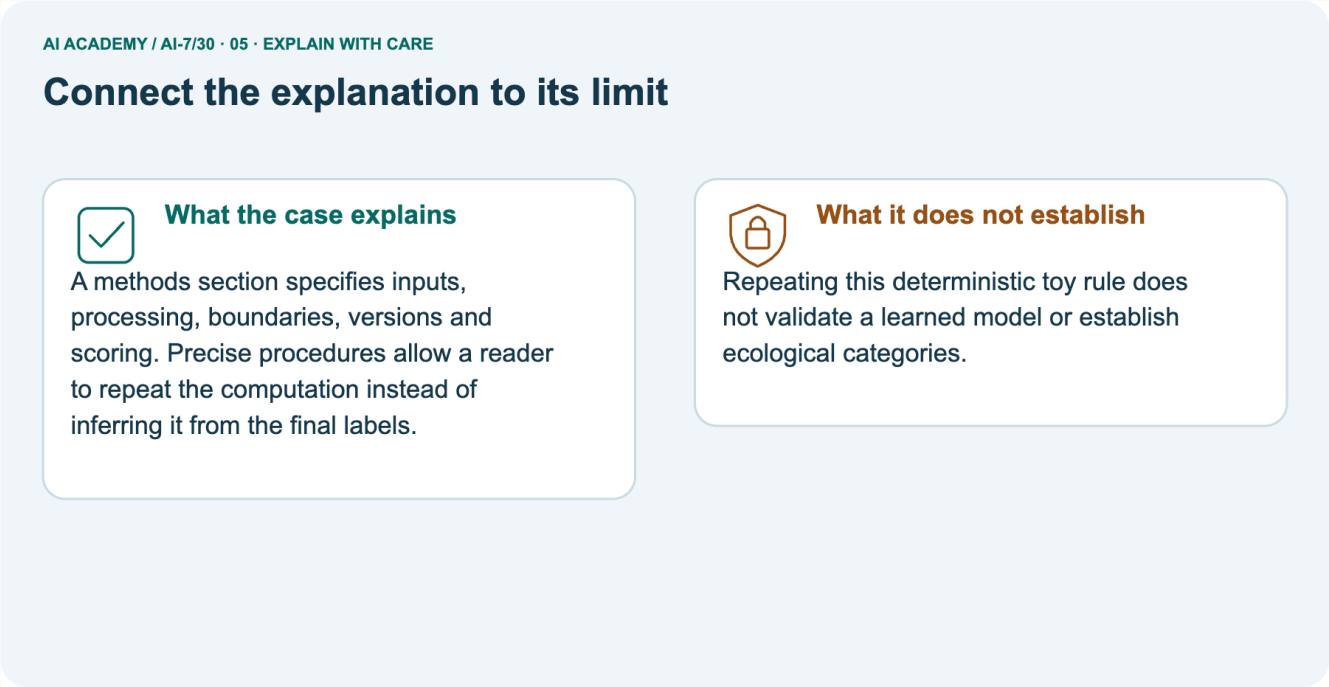
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Oak is short; pine and willow are long. Changed condition: Change the rule to length greater than 4. Expected result: Pine becomes short; willow stays long and oak stays short.

Reason: The equality case distinguishes the two procedures.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



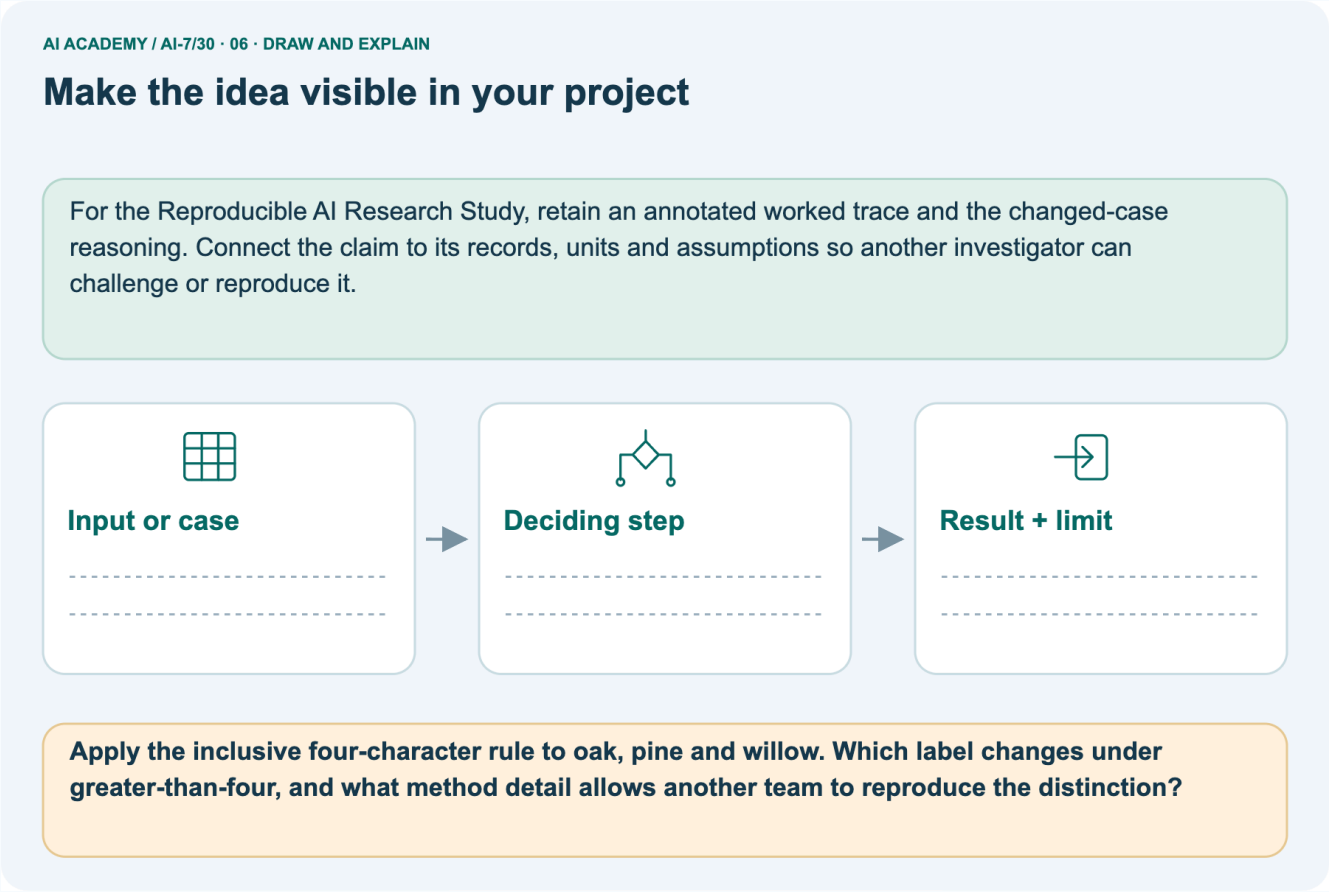
Follow the labelled picture with the matching explanation. Open the vector diagram

A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Limit: Repeating this deterministic toy rule does not validate a **learned model** or establish ecological categories.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Writing the methods section: the case

02. Follow the deciding steps

03. Writing the methods section — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.